

Notes: Greenwood, Hanson, and Stein (JF, 2015)

A Comparative-Advantage Approach to Government Debt Maturity

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1 Big picture

- **Question.** How should a high-credit-quality government choose the maturity structure of its debt when short-term safe debt provides monetary services to investors?
- **Core insight.** Treasury bills are not just low-risk assets in the standard asset-pricing sense. They provide *short-term safety* and therefore embed a convenience premium. This gives the government a reason to borrow short, because short-term debt can be socially valuable even if it raises rollover risk.
- **Government-only trade-off.** In the baseline model, the government is the only producer of riskless money-like claims. It balances:
 - the **money-premium benefit** of issuing more short-term bills; and
 - the **rollover-risk cost** of having to refinance debt more often when future short rates are uncertain.
- **Why the paper matters.** Standard tax-smoothing models tend to push the government toward long-term debt. This paper shows that once one recognizes the convenience yield on short-term safe assets, there is a clear reason for an interior maturity structure with meaningful short-term issuance.
- **Main positive result.** In the government-only model, the optimal short-term share is larger when money demand is stronger and smaller when rollover risk is larger. It is also smaller when the government's overall debt burden is larger, because refinancing risk matters more in dollar terms.
- **A key conceptual clarification.** It is not enough that short-term debt has a lower expected return in a standard risk-premium sense. The relevant reason to borrow short is that short-term debt offers monetary services, not that it has a low beta or a low term premium for ordinary asset-pricing reasons.
- **Private-sector extension.** Financial intermediaries can also create short-term safe claims by funding risky long-term assets with short-term debt. But private money creation can be excessive because intermediaries do not fully internalize the fire-sale externalities associated with short-term funding.
- **Comparative-advantage result.** If private money creation has negative externalities, the government should issue more short-term bills and partially crowd out the private sector. The government has a comparative advantage in producing safe short-term paper when its marginal social cost of doing so is lower than the private sector's fire-sale cost.
- **Regulatory interpretation.** Government debt management becomes a complement to financial regulation. If direct regulation of private short-term funding is costly, incomplete, or easy to arbitrage around, issuing more T-bills can reduce private incentives to create fragile money-like claims.
- **Empirical motivation.** The paper documents three stylized facts that line up with the theory:
 - the shortest T-bills carry especially large convenience premia;

- increases in T-bill supply reduce that premium;
- increases in T-bill supply are associated with less private short-term money creation.
- **Calibration message.** The estimated crowding-out benefit of short-term Treasury issuance is plausibly of the same order of magnitude as the direct convenience benefit from bills themselves.
- **Big takeaway.** Debt management is not just about minimizing financing costs or smoothing taxes. It also shapes the economy-wide supply of safe short-term assets and therefore interacts directly with financial stability.

2 Data and measurement (key objects)

This paper mixes **stylized-fact evidence** with a **normative model**. So the relevant objects are both empirical series and model variables.

2.1 Convenience-premium data for Treasury bills

The paper measures the money premium on short-term bills using **z-spreads**, defined as the difference between the actual yield on an n -week Treasury bill and the fitted yield from a smooth Treasury yield curve estimated from notes and bonds with remaining maturity above three months.

Formally,

$$z_t^{(n)} \equiv y_t^{(n)} - \hat{y}_t^{(n)}.$$

Interpretation. A more negative z-spread means the bill yield is unusually low relative to the rest of the yield curve, which the paper interprets as a stronger convenience premium on short-term safety.

The main bill-yield sample runs from **1983 to 2009**. The fitted yields come from the G"urkaynak–Sack–Wright yield curve procedure.

2.2 Government supply variables

The empirical work separates Treasury supply into:

- **BILLS/GDP:** the ratio of Treasury bills to GDP,
- **NONBILLS/GDP:** the ratio of Treasury notes and bonds to GDP.

These are constructed from detailed Treasury auction data so the supply of short-term and long-term government debt can be tracked at weekly frequency.

Interpretation. This decomposition is central because the theory predicts that *short-term* government debt should matter more for money premia and crowding out than long-term debt.

2.3 Private money creation

The paper uses **financial commercial paper** as a direct, high-frequency proxy for private money creation:

$$\text{FINCP}/\text{GDP}.$$

The data are available:

- weekly from **2001–2009**,
- monthly from **1992–2009**,
- quarterly from **1952–2009**.

Interpretation. Financial commercial paper is not all private money creation, but it is one of the cleanest short-term safe-like private claims observed at high frequencies.

2.4 Debt maturity and debt burden

To study how government maturity behaves over time, the paper uses:

- the **weighted-average maturity of marketable Treasury debt** from CRSP issue-level data,
- the **debt-to-GDP ratio** from historical Treasury debt data.

The maturity sample runs from **1952 to 2009**.

Interpretation. This allows the authors to compare one of the model’s main comparative statics—higher debt burden implies longer optimal maturity—to U.S. time-series evidence.

2.5 Key model objects

The model revolves around the following objects:

- β : the random date-1 discount factor for date-2 consumption,
- M_0 : money-like short-term safe debt held by households,
- D : total government debt issued at date 0,
- S : the short-term share of government debt, so $M_G = SD$,
- M_P : private money creation by intermediaries,
- k : the fire-sale discount factor on bank assets,
- ϕ : the fraction of project returns that private investors can appropriate,
- $g(K)$: the production technology for patient investors in the crisis state.

The paper also defines

$$b \equiv \mathbb{E} \left[\frac{(\beta - 1)^2}{1 + \beta} \right] \approx \frac{\text{Var}(\beta)}{2},$$

which summarizes the magnitude of rollover risk.

Interpretation. b is the reduced-form statistic that tells you how painful it is for the government to rely on short-term refinancing.

2.6 Two kinds of safety

A subtle but crucial measurement idea in the paper is the distinction between:

- **short-term safety:** near-term stability of market value, especially relevant for T-bills;
- **long-term safety:** certainty of ultimate repayment, more relevant for long-term Treasuries.

Interpretation. Because financial intermediaries compete with the Treasury mostly in supplying *short-term* safety, changing the maturity mix of government debt is more important than changing only the total quantity of debt.

3 Models and empirical specifications (all equations + interpretation)

3.1 Empirical stylized-fact regressions

The first empirical specification links bill convenience premia to the supply of Treasury bills:

$$z_t^{(n)} = a^{(n)} + b^{(n)} (\text{BILLS}/\text{GDP})_t + c^{(n)} (\text{NONBILLS}/\text{GDP})_t + d^{(n)} t + \varepsilon_t^{(n)}. \quad (1)$$

The paper also estimates this regression in four-week changes:

$$\Delta_4 z_t^{(n)} = a^{(n)} + b^{(n)} \Delta_4 (\text{BILLS}/\text{GDP})_t + c^{(n)} \Delta_4 (\text{NONBILLS}/\text{GDP})_t + \varepsilon_t^{(n)}. \quad (2)$$

Interpretation. A positive $b^{(n)}$ means that when the Treasury supplies more bills, the z-spread rises, so the bill convenience premium shrinks.

To study crowding out of private money creation, the paper estimates

$$(\text{FINCP}/\text{GDP})_t = a + b (\text{BILLS}/\text{GDP})_t + c (\text{NONBILLS}/\text{GDP})_t + dt + u_t, \quad (3a)$$

$$\Delta_k (\text{FINCP}/\text{GDP})_t = a + b \Delta_k (\text{BILLS}/\text{GDP})_t + c \Delta_k (\text{NONBILLS}/\text{GDP})_t + \Delta_k u_t. \quad (3b)$$

Interpretation. A negative b means that more short-term Treasury issuance crowds out private short-term funding.

Finally, to see whether debt maturity actively adjusts to the debt burden, the paper estimates a partial-adjustment equation for the maturity of new issuance:

$$M_t^{\text{Iss}} = 0.98 + 2.86 (D/\text{GDP})_t + 0.76 M_{t-1}^{\text{Iss}} - 0.34 M_{t-1}^{\text{out}}, \quad R^2 = 0.55. \quad (4)$$

Interpretation. When debt-to-GDP is high, the Treasury tends to lengthen the maturity of new issuance.

3.2 The government-only model

Households consume across three dates and value money-like short-term safe debt. Their utility is

$$U = C_0 + \mathbb{E}[C_1 + \beta C_2] + v(M_0), \quad (5)$$

where $M_0 = B_{0,1}$ is the quantity of short-term government bonds held at date 0.

Interpretation. The term $v(M_0)$ is the convenience value of short-term safe debt. This is what generates a reason for short maturity that is absent from standard tax-smoothing models.

The government budget constraints are

$$\begin{aligned} G &= \tau_0 + B_{0,1}P_{0,1} + B_{0,2}P_{0,2}, \\ B_{0,1} &= \tau_1 + B_{1,2}P_{1,2}, \\ B_{1,2} + B_{0,2} &= \tau_2. \end{aligned} \quad (6)$$

Household consumption is therefore

$$\begin{aligned} C_0 &= 1 - \tau_0 - \frac{1}{2}\tau_0^2 - B_{0,1}P_{0,1} - B_{0,2}P_{0,2}, \\ C_1 &= 1 - \tau_1 - \frac{1}{2}\tau_1^2 + B_{0,1} - B_{1,2}P_{1,2}, \\ C_2 &= 1 - \tau_2 - \frac{1}{2}\tau_2^2 + B_{1,2} + B_{0,2}, \end{aligned} \quad (7)$$

and after substituting in the government budget constraints,

$$\begin{aligned} C_0 &= 1 - \frac{1}{2}\tau_0^2 - G, \\ C_1 &= 1 - \frac{1}{2}\tau_1^2, \\ C_2 &= 1 - \frac{1}{2}\tau_2^2. \end{aligned} \quad (8)$$

Interpretation. Taxes are distortionary, and the deadweight loss is quadratic in taxes. This creates the usual tax-smoothing motive.

The planner's baseline problem is

$$\max_{\{B_{0,1}, B_{0,2}, B_{1,2}\}} v(B_{0,1}) - \frac{1}{2} \left(\tau_0^2 + \mathbb{E}[\tau_1^2] + \mathbb{E}[\beta \tau_2^2] \right). \quad (9)$$

Without money demand, this reduces to

$$\min_{\{B_{0,1}, B_{0,2}, B_{1,2}\}} \frac{1}{2} \left(\tau_0^2 + \mathbb{E}[\tau_1^2] + \mathbb{E}[\beta \tau_2^2] \right). \quad (10)$$

At date 1, optimal rollover satisfies

$$B_{1,2} = \frac{B_{0,1} - B_{0,2}}{1 + \beta}. \quad (12)$$

Interpretation. The government uses date-1 borrowing to smooth taxes between dates 1 and 2.

Proposition 1. Without money demand, the government perfectly smooths taxes:

$$\tau_0 = \tau_1 = \tau_2 = \frac{G}{3}, \quad B_{0,1} = B_{0,2} = \frac{G}{3}, \quad B_{1,2} = 0.$$

Economic meaning. In the benchmark with no convenience yield, there is no reason to bias issuance toward the short end. The government can simply lock in long-term financing and equalize tax distortions across dates.

3.3 Optimal maturity with money demand

Define total date-0 government debt and the short-term debt share by

$$D = B_{0,1} + B_{0,2}, \quad S = \frac{B_{0,1}}{D}.$$

Then the planner's problem can be rewritten as

$$\min_{S,D} \frac{1}{2}(G - D - DSv'(DS))^2 + \frac{D^2}{2} \left[b \left(S - \frac{1}{2} \right)^2 + \frac{1}{2} \right] - v(DS). \quad (13)$$

The first-order condition for the short-term share is

$$Db \left(S - \frac{1}{2} \right) = v'(SD) + \tau_0 \left(v'(SD) + SDv''(SD) \right). \quad (14)$$

For most of the paper the authors suppress the tax-lowering term and focus on the cleaner condition

$$Db \left(S - \frac{1}{2} \right) = v'(SD). \quad (14')$$

Interpretation. The left-hand side is the marginal rollover-risk or tax-smoothing cost of shortening maturity. The right-hand side is the marginal money benefit of short-term debt.

Proposition 2. The optimal short-term share satisfies $S^* > 1/2$. Moreover:

- $\partial S^* / \partial b < 0$: more refinancing risk means longer optimal maturity;
- $\partial S^* / \partial G < 0$: a higher debt burden means longer optimal maturity;
- if money demand is stronger, S^* rises.

Economic meaning. The government borrows short because short-term debt is socially valuable, but it does so less aggressively when the consequences of rollover risk are larger.

3.4 Foreign holders and seignorage

The paper also studies a nationalistic-planner case where the tax-lowering benefit of short-term debt is emphasized. The planner's objective becomes

$$\min \frac{1}{2} \left(\tau_0^2 + \mathbb{E}[\tau_1^2] + \mathbb{E}[\beta \tau_2^2] \right) - v'(M)M. \quad (15)$$

Interpretation. If foreigners hold the short-term debt, then the lower financing cost looks like seignorage revenue extracted from abroad. The paper shows that the comparative statics remain qualitatively similar: money demand still pushes the government toward shorter maturities, while larger debt and more rollover risk push it toward longer maturities.

3.5 Banks and private money creation

Banks invest I at date 0 in risky long-term assets. If bad news arrives at date 1, short-term creditors run and banks must sell a fraction Δ of assets in a fire sale. The fire-sale discount is pinned down by the patient investors' outside option:

$$\frac{1}{k} = \phi g'(K) = \phi g'(W - M_P). \quad (19)$$

Interpretation. More private money creation (M_P) leaves patient investors with less cash for new investment in bad states, raises the marginal value of capital, and therefore worsens fire-sale losses.

A bank that finances itself with private money M_P solves

$$\Pi = [pF + (1-p)\lambda I - I] + M_P \left(v'(M_0) - (1-p)\left(\frac{1}{k} - 1\right) \right). \quad (20)$$

The equilibrium quantity of private money satisfies

$$v'(M_P^* + M_G) = (1-p) \left(\phi g'(W - M_P^*) - 1 \right). \quad (21)$$

Interpretation. Private intermediaries issue short-term safe debt until the marginal convenience benefit equals their *private* expected fire-sale cost. Because only $\phi < 1$ of the return is pledgeable, they underweight the social cost of fire sales.

3.6 The social planner with government and private money

Social welfare is

$$U^{\text{SOCIAL}} = \mathbb{E}[g(K) - K] + v(M_0) - \frac{1}{2} \left(\tau_0^2 + \mathbb{E}[\tau_1^2] + \mathbb{E}[\beta \tau_2^2] \right). \quad (22)$$

If the planner can directly control private money creation, the first-best condition is

$$(1-p)(g'(W - M_P^{**}) - 1) = v'(M_G^{**} + M_P^{**}) = b \left(M_G^{**} - \frac{D^{**}}{2} \right). \quad (23)$$

Interpretation. In the first best, the planner equalizes the marginal social benefit of additional money services to the marginal social cost of both public and private money creation.

When the government cannot directly regulate private money creation, it relies on crowding out. The private-sector reaction function satisfies

$$-1 < \frac{\partial M_P^*}{\partial M_G} = - \frac{v''(M_P^* + M_G)}{v''(M_P^* + M_G) + (1-p)\phi g''(W - M_P^*)} < 0. \quad (24)$$

The second-best government condition becomes

$$D^{***}b\left(S^{***} - \frac{1}{2}\right) = v'(M_P^* + S^{***}D^{***}) + (1-p)(\phi-1)g'(W - M_P^*)\frac{\partial M_P^*}{\partial M_G}. \quad (25)$$

Interpretation. Relative to the government-only model, short-term government debt now has *two* benefits:

- it directly provides monetary services;
- it indirectly lowers private money creation by shrinking the convenience premium.

Proposition 5. If private money creation generates externalities ($\phi < 1$), then the government issues more short-term debt than it would if it ignored crowding out.

Proposition 6. If the government also has a direct regulatory tool, such as a Pigouvian tax on private money creation, then debt management and regulation are used together. The dirtier or harder regulation is to implement, the more important debt management becomes.

Proposition 7. With more than two maturities, the government may want a **barbell** structure: more very short and very long debt, less medium-term debt. If yield-curve shocks are mainly parallel shifts, this helps preserve money creation while limiting tax volatility. If the yield curve also changes shape, the barbell logic becomes weaker.

4 Identification and instruments (what makes the estimates credible)

4.1 What the theory is trying to isolate

The theory is deliberately stripped down. The government is assumed to be free of default risk, endowments are fixed, and there are no fiscal shocks. These assumptions remove classic hedging motives for state-contingent public debt and isolate the paper's new mechanism: **a convenience-yield motive for short-term debt versus a rollover-risk motive for long-term debt.**

Why this helps. It makes the paper's logic clean. If the government chooses shorter maturity in the model, it is because short-term safe debt provides monetary services or crowds out fragile private money creation—not because of hidden default-risk, inflation-risk, or hedging channels.

4.2 Endogeneity in the money-premium regressions

A central concern is that Treasury bill supply is not randomly assigned. If money demand rises, the Treasury may itself respond by issuing more bills. This is especially visible in late 2008, when bill supply jumped just as money demand surged.

Why OLS is problematic. In the regression of z-spreads on bill supply, endogenous Treasury issuance tends to bias the OLS estimate downward. The government partially accommodates the demand shock, which dampens the observed relationship between supply and convenience premia.

4.3 The instrumental-variables strategy

The paper addresses this concern with an IV based on **seasonal tax-calendar variation**. Ahead of major tax dates, the Treasury temporarily expands bill issuance to manage its cash balances and then repays this borrowing after receipts arrive.

Why the instrument is plausible.

- The timing is driven by statutory tax deadlines rather than by weekly changes in convenience yield.
- The induced variation is concentrated in very short-term T-bill supply, exactly where the paper expects the strongest pricing effects.
- The first-stage relationship is strong, with bill supply showing large and systematic spikes around tax dates.

Empirical implication. The IV estimates are larger than the OLS estimates, which is exactly what one would expect if the Treasury endogenously offsets money-demand shocks.

4.4 Why comparing bills to nonbills matters

The paper does not ask only whether more government debt matters. It asks whether **short-term** government debt matters more than **long-term** government debt. That is why the regressions always separate BILLS/GDP from NONBILLS/GDP.

Why this is important. The theory is about short-term safety. If long-term Treasuries behaved the same way as bills, the comparative-advantage argument would be much weaker.

4.5 Identification of private crowding out

The evidence on crowding out is more suggestive than the evidence on convenience premia. The paper studies financial commercial paper because it is a direct, observable form of private short-term money creation, but it is not the whole universe of private money.

What strengthens the case.

- The coefficients on bill supply are consistently negative across many samples and frequencies.
- The estimated crowding-out effect grows in magnitude at longer horizons, which fits the idea that quantities adjust more slowly than prices.
- The VAR evidence shows an immediate pricing response and a slower quantity response, matching the paper's mechanism.

Main limitation. The paper cannot claim a perfectly clean causal estimate of total private-money crowding out. It only shows that the observed piece of private money creation moves in the right direction and with economically meaningful magnitude.

4.6 Debt maturity and debt-to-GDP evidence

The strong positive time-series correlation between government debt maturity and debt-to-GDP is best read as a **stylized-fact validation** of the model’s comparative statics, not as a causal design.

Why it still matters. The theory predicts exactly this pattern: when the stock of debt is larger, rollover risk becomes more painful, so the government should choose longer maturity.

4.7 Calibration credibility

The quantitative calibration is transparent rather than overengineered. The paper assumes:

- crisis probability $(1 - p) = 5\%$,
- nonpledgeability $(1 - \phi) = 15\%$,
- marginal crisis-state social return $g'(\cdot) = 1.30$,
- crowding-out elasticity $\partial M_P^* / \partial M_G \approx -0.50$.

Interpretation. These inputs are used to get an *order-of-magnitude* result—that the crowding-out benefit is about 50 bps, roughly comparable to the direct convenience yield on bills. The calibration is persuasive as a policy back-of-the-envelope, but it is not intended as a tightly estimated structural exercise.

4.8 Bottom line on credibility

The paper is strongest when read as a combination of:

- a clean normative model,
- empirical stylized facts that line up with the key premises,
- and a calibration showing the mechanism is plausibly first order.

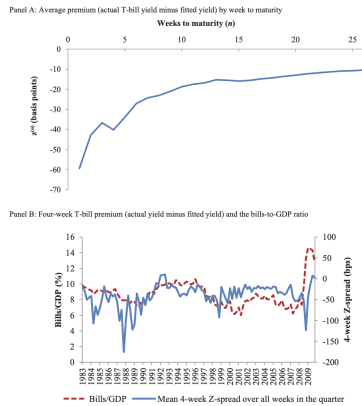
Its main weakness is that the private crowding-out evidence is not as airtight as the convenience-premium evidence. But the overall theory-and-evidence package is coherent and internally consistent.

5 Tables and figures

5.1 Figure 1 and Table I: short-term Treasuries carry a special premium

Read:

- Figure 1 shows that the shortest T-bills trade at yields far below what the rest of the Treasury curve would predict.
- The average z-spread is about -40 bps for four-week bills and roughly -60 bps for one-week bills.



- Panel B of Figure 1 shows that when bills as a share of GDP rise, the four-week z-spread rises too—that is, the convenience premium shrinks.
- Table I confirms this in regressions. In the full 1983–2009 sample, a one-percentage-point increase in bills-to-GDP raises the two-week z-spread by about 5.8 bps, and the effect is larger before the crisis period.

Economic message. Short-term bills are special. The shortest maturities embed the largest money premium, which is exactly the raw fact the model needs.

5.2 Figure 2: seasonal bill supply around tax dates

Read:

- Figure 2 shows large, predictable spikes in bill issuance around major federal tax deadlines.
- These spikes reverse after tax receipts arrive.
- This seasonal pattern provides the instrument used in the high-frequency pricing regressions.

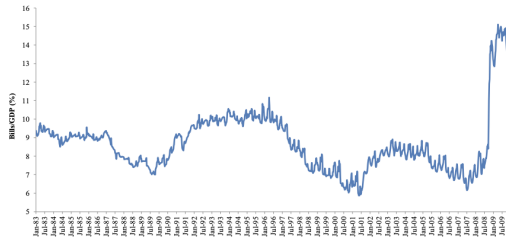
Economic message. The IV strategy is grounded in a transparent institutional feature of Treasury cash management, not in a black-box instrument.

5.3 Table II: government bills crowd out private short-term funding

Read:

- Table II shows that increases in bill supply are associated with declines in financial commercial paper issuance.
- The average crowding-out coefficient is about -0.14 in samples ending in 2009 and about -0.24 in samples ending in 2007.
- In the monthly 1992–2007 sample, the coefficient becomes substantially larger in long-horizon

Panel A: The ratio of bills to GDP from 1983 to 2009



Panel B: High-frequency variation in the supply of bills

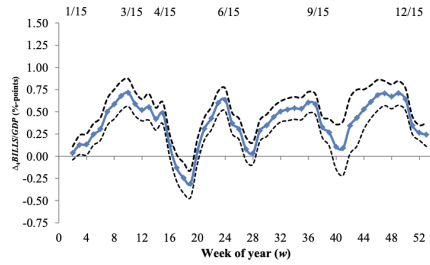


Figure 2. The supply of Treasury bills, 1983–2009. Panel A plots the ratio of Treasury bills to GDP each week from 1983 to 2009. To compute the ratio of Treasury bills to GDP precisely at the end of each week, we use detailed data on the size and timing of Treasury auctions. Panel B plots the coefficients from the first-stage regression from our instrumental variables estimator for 1983–2009. Specifically, we regress $\Delta_4 \text{BILLS}/\text{GDP}$ on a full set of week-of-year dummies (the omitted category is the first week of the year):

$$\Delta_4(\text{BILLS}/\text{GDP})_t = c + \sum_{w=2}^{53} d^{(w)} \cdot \mathbf{1}\{\text{week}(t) = w\} + \Delta_4 v_t.$$

Confidence intervals are based on Newey-West (1987) standard errors allowing for serial correlation up to eight weeks. We mark the six major Federal tax deadlines on the figure: January 15, March 15, April 15, June 15, September 15, and December 15.

differences, reaching roughly -0.53 for annual changes.

Economic message. The quantity evidence is weaker than the pricing evidence, but it points in the same direction: when the government supplies more short-term safe paper, the private sector supplies less.

5.4 Figure 3: prices move fast, quantities move more slowly

Read:

- The VAR impulse responses show that z-spreads respond immediately to a shock in bill supply.
- The response in financial commercial paper takes longer, with the quantity adjustment unfolding over months rather than days.
- In the shorter 1992–2007 sample, the quantity effect appears faster; in the longer 1983–2007 sample, it is more gradual.

Economic message. This dynamic pattern matches the theory nicely: convenience premia reprice quickly, while the private sector’s funding mix adjusts more slowly.

5.5 Figure 4: debt burden and debt maturity move together

Read:

- Figure 4 plots the weighted-average maturity of U.S. Treasury debt against debt-to-GDP from 1952 to 2009.

Table II
Financial Commercial Paper and the Supply of Short-Term Treasuries

This table reports regressions of financial commercial paper supply on the supply on Treasury bills scaled by GDP and the supply of Treasury notes and bonds: $FPCP_t/\text{GDP}_t = \alpha + \beta_1(\text{BILLS}/\text{GDP})_t + \beta_2(\text{NOTES}/\text{GDP})_t + d_t + u_t$ and $\Delta_4(FPCP/\text{GDP})_t = \alpha + \beta_1 \Delta_4(\text{BILLS}/\text{GDP})_t + \beta_2 \Delta_4(\text{NOTES}/\text{GDP})_t + \Delta_4 u_t$. Weekly data on outstanding unsecured financial commercial paper are available from the Federal Reserve starting in 2001, monthly data are available from the Federal Reserve starting in 1992, and quarterly data are available from Table I106 of the Flow of Funds starting in 1952. To maintain comparability with the recent weekly and monthly data, we use open market paper issued by “domestic financial businesses” less open market paper issued by “ABS issuers” in Table I. 2008.3 To compute the ratio of Treasury bills, notes, and bonds to GDP at the end of each week and month, we use detailed data on the size and timing of Treasury auctions from <http://www.treasurydirect.gov>. For the quarterly 1952–2009 sample, we use data from the Monthly Statement of Public Debt to construct the supply of Treasury bills and Treasury notes and bonds. We include marketable Treasury certificates (interest-bearing issues with original maturities less than one year) that the Treasury issued until 1967 in our bills measure. We estimate these regressions in levels and changes computed at different horizons. The t -statistics are shown in brackets. For the levels regressions, we compute standard errors assuming that the residuals follow an AR(1) process. For the changes regressions, we compute Newey-West (1987) standard errors. For each specification, the table lists the number of lags used in computing Newey-West standard errors. For the bivariate regressions, we report the t -statistic for the hypothesis that $\beta_1 - \beta_2 = 0$, labeled $t_{\beta_1 - \beta_2}$, below. The IV specifications for four-week changes in columns (5) and (6) use the same methodology as in Table I. Panel A shows results for samples ending in 2009; Panel B shows results for samples ending in 2007.

	Weekly (2001+)						Monthly (1992+)						Quarterly (1952+)					
	Levels		4-week changes		4-week changes		Levels		1-month changes		3-month changes		12-month changes		4-quarter changes			
	OLS	OLS	OLS	IV	OLS	IV	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	IV		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	
Panel A: Samples Ending in 2009																		
β	-0.174	-0.150	-0.139	-0.139	-0.055	-0.055	-0.165	-0.299	-0.060	-0.064	-0.115	-0.109	-0.210	-0.216	-0.116	-0.211	-0.087	
β_1	[-2.79]	[-2.09]	[3.21]	[-3.29]	[-1.46]	[-1.41]	[-2.20]	[-3.75]	[-1.39]	[-1.47]	[-6.48]	[-5.53]	[-5.94]	[-2.88]	[-1.17]	[-3.22]	[-4.09]	
β_2	-0.004	-0.001	-0.001	-0.002	0.001	0.001	0.001	0.001	0.001	0.001	-0.025	-0.025	0.009	0.008	0.048	0.048		
β_3	[-0.50]	[-0.03]	[-0.03]	[-0.37]	[0.80]	[0.80]	[0.65]	[0.65]	[0.65]	[0.65]	[-0.69]	[-0.69]	[0.15]	[0.15]	[1.32]	[1.32]		
d	0.397	0.321			-0.018	-0.019									0.021	0.021		
d_1	[0.35]	[0.52]			[-3.12]	[-1.02]									[0.84]	[0.84]		
N	469	469	465	465	465	465	214	213	213	213	211	211	202	202	232	228		
SE	481	481	NW 8	NW 8	NW 8	NW 8	481	481	NW 4	NW 4	NW 4	NW 4	NW 4	NW 4	481	481		
R^2	0.42	0.43	0.14	0.14	0.56	0.66	0.03	0.03	0.12	0.13	0.45	0.45	0.74	0.76	0.11	0.15		
$F_{\beta_1 - \beta_2}$			[-0.93]	[-2.60]		[-0.40]		[-3.48]		[-1.54]		[-1.84]		[-1.80]		[-1.43]		

(Continued)

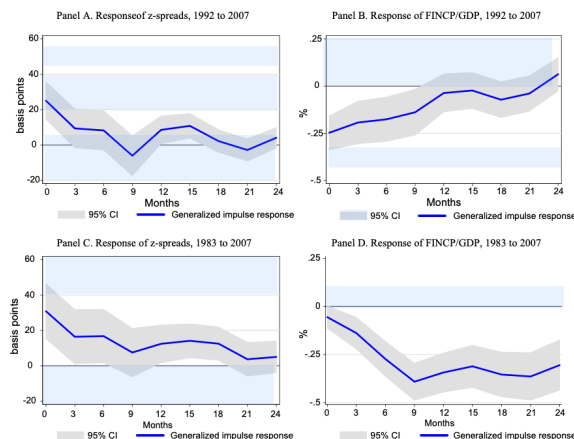


Figure 3. Impulse responses to shocks in the supply of T-bills. We estimate a vector autoregression (VAR) using monthly data from 1992 to 2007 (Panels A and B) and 1983 to 2007 (Panels C and D). We run a VAR in levels that includes *BILLS/GDP*, *NONBILLS/GDP*, *FINCP/GDP*, the *z*-spread on four-week T-bills at 3-, 6-, 9-, and 12-month lags, as well as a deterministic time trend. Below we plot the estimated generalized impulse response functions (Pesaran and Shin (1998)) for *z*-spreads and *FINCP/GDP* following a shock to *BILLS/GDP*. To make the magnitudes comparable to those in Tables I and II, we plot the response functions corresponding to a one-percentage-point impulse to *BILLS/GDP*. For the bottom two panels, the underlying quarterly series for *FINCP/GDP* is interpolated between quarter-ends.

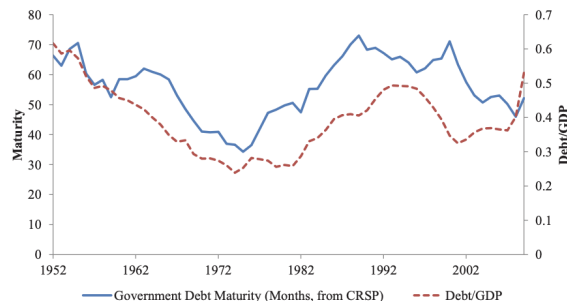


Figure 4. Debt-to-GDP and the maturity of U.S. Treasury debt, 1952–2009. The figure plots the weighted-average maturity of all marketable Treasury debt against the debt-to-GDP ratio from 1952 to 2009. The weighted average maturity is calculated using issue-level data on debt outstanding from CRSP and excludes data on Treasury Inflation Protected Securities (TIPS). The debt-to-GDP ratio is obtained from Henning Bohn's website.

- The correlation is strongly positive, about 0.71.
- The maturity of new issuance also responds positively to the debt burden in the partial-adjustment regression.

Economic message. This is the cleanest reduced-form validation of the government's rollover-risk trade-off: when there is more debt to refinance, the Treasury lengthens maturity.

5.6 The calibration in Section III.E

Read:

- Using a 5% crisis probability, 15% nonpledgeability, a crisis-state marginal return of 1.30, and a crowding-out elasticity of -0.50 , the paper estimates a crowding-out benefit of about 50 bps.
- The direct money benefit from short-term bills is around 40 bps.
- Taking the crowding-out benefit seriously could justify increasing the Treasury's short-term share by roughly 4 percentage points, and perhaps more under alternative parameter choices.

Economic message. The crowding-out channel is not a tiny second-order refinement. On the paper's own numbers, it is comparable to the direct convenience-yield channel.

5.7 The policy extension with multiple maturities

Read:

- If yield-curve shocks are mostly parallel, a barbell strategy can create lots of short-term safe assets while limiting the government's effective interest-rate exposure.
- If nonparallel shifts matter, the hedge is imperfect and the government must accept some shortening of effective maturity to create more money.

Economic message. The policy prescription is not mechanically *issue only bills*. It is *use the maturity structure strategically*, recognizing both convenience services and risk management.

6 Short summary

Greenwood, Hanson, and Stein study government debt maturity when short-term safe debt provides monetary services. In their baseline model, the government faces a clean trade-off: issuing more T-bills lowers financing costs because bills carry a convenience premium, but it also raises rollover risk by exposing future taxes to uncertain refinancing rates. This yields an interior optimum and predicts that the government should shorten maturity when money demand is stronger and lengthen maturity when debt-to-GDP or rollover risk is higher.

The paper's most important extension adds private money creation by financial intermediaries. Because banks can issue short-term safe-looking claims backed by risky long-term assets, they compete with the government in the production of money-like securities. But banks do not internalize the full fire-sale costs of this activity. As a result, government issuance of short-term bills can play a regulatory role by shrinking the convenience premium and crowding out socially excessive private money creation. The empirical stylized facts and calibration both suggest this channel is quantitatively meaningful.

7 Conclusion

Core conclusion. Optimal government debt maturity depends not only on tax smoothing, but also on the monetary services of short-term safe debt and on the government's comparative advantage relative to the private sector in producing such debt.

Economic intuition. Treasury bills are socially useful because investors value short-term safety. But when private intermediaries create the same sort of short-term safety, they do so in a fragile way that can trigger fire sales. The government can partially replace that fragile private money with public money.

Policy implication. Debt management should be thought of as part of the financial-stability toolkit. When direct regulation of private money creation is incomplete or costly, shortening government maturity can reduce systemic fragility by crowding out private short-term funding.

Takeaway.

- The paper's central condition is a **marginal trade-off**: money-premium benefits versus rollover-risk costs.
- The distinctive extension is the **comparative-advantage** logic: the government may want to issue more bills not only because bills are cheap, but because doing so displaces fragile private money creation.
- Empirically, the key supporting facts are: short T-bills are special, bill supply moves their convenience premium, and more bill supply is associated with less private short-term funding.